SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Real-Time Data Analytics Enabled Smart Vehicle Rental Management System for Enhanced Decision Support

CSA0927-Programming in Java for Problem Solving

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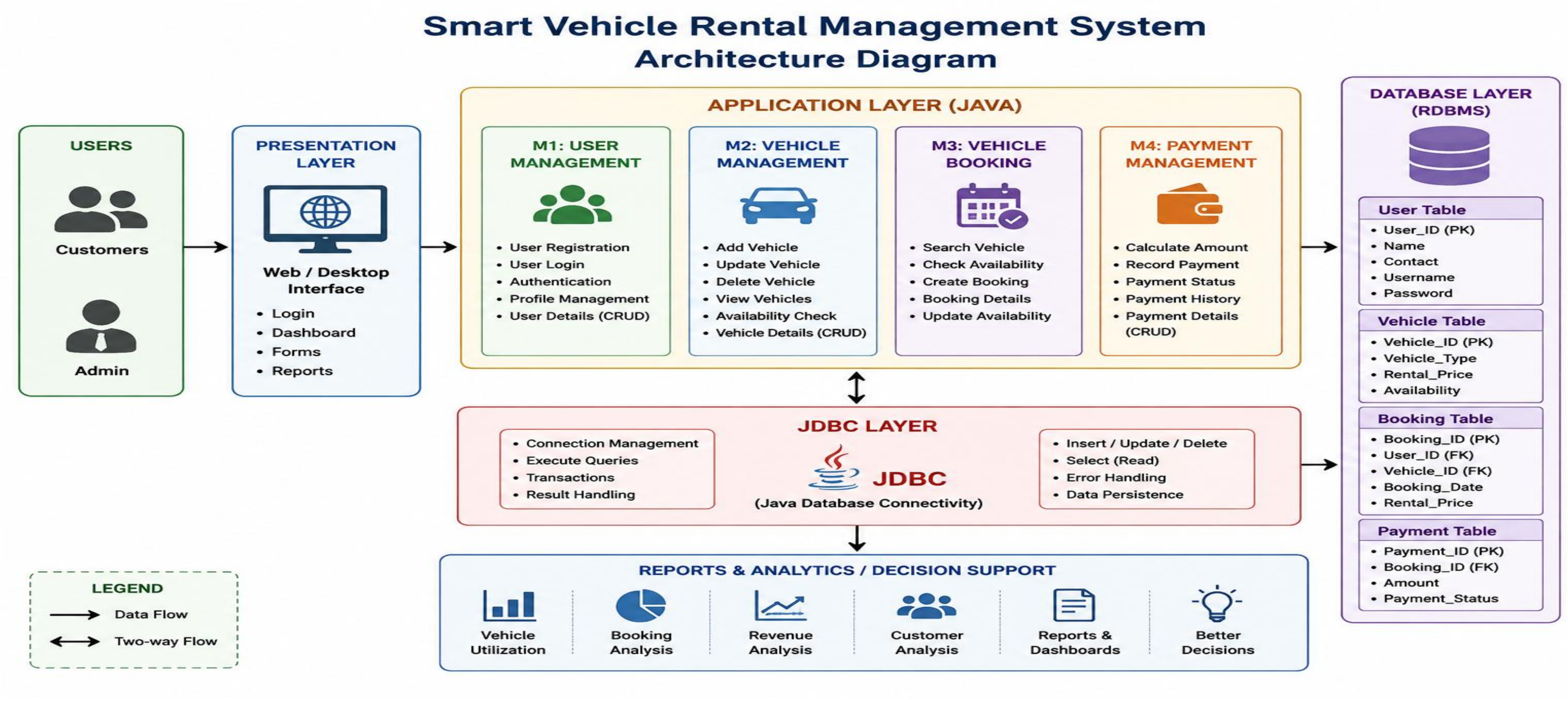
Introduction

- Vehicle rental services require efficient management of users, vehicles, bookings, and payments.
- Traditional rental management can involve manual records and delayed information.
- The proposed system provides a Java-based smart vehicle rental management platform.
- It manages vehicle details, customer bookings, rental prices, and payment status.

- Real-Main Idea
- Manage → Book → Pay → Store Data → Analyze → Support Decisionstime information can help administrators make better rental and business decisions.

Architecture Diagram



Literature Review

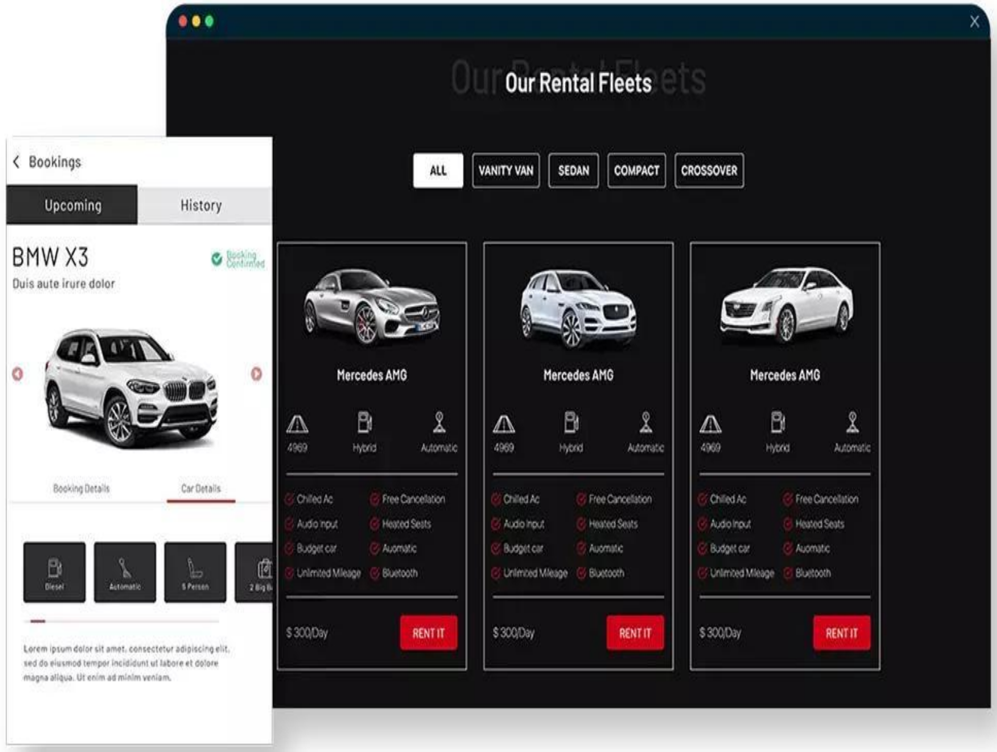
Author / Year	Title / Source	Key Contribution	Limitations / Gaps
Sharma et al. (2025)	Automated Car Rental Management System Using Java	Developed a Java-based system for vehicle booking and rental management.	Limited support for mobile access and online payments.
Patel and Mehta (2024)	Online Vehicle Rental System with Customer Management	Improved customer registration, booking, and rental tracking.	Lacks real-time vehicle tracking features.
Verma et al. (2024)	Java-Based Car Reservation System	Implemented automated reservation and availability management.	Does not include advanced security mechanisms.
Reddy and Rao (2023)	Vehicle Rental Management Using Object-Oriented Programming	Applied OOP concepts for efficient rental operations.	Limited scalability for large rental businesses.

Problem Statement

- Traditional vehicle rental management systems may face problems such as:
- Manual management of customer information.
- Difficulty tracking vehicle availability.
- Booking conflicts.
- Inefficient payment management.
- Difficulty maintaining rental records.
- Lack of centralized database management.
- Limited information for decision-making.

Objectives

- Main Objective
- To develop a Java-based smart vehicle rental management system for efficient management of users, vehicles, bookings, and payments.
- Specific Objectives
- Provide secure user login.
- Manage user details.
- Manage vehicle information.
- Display vehicle availability.
- Allow vehicle booking.
- Manage rental prices..



Engineer to Excel

Proposed System

User
↓
Login
↓
Vehicle Management
↓
Vehicle Availability
↓
Booking
↓
Rental Price Calculation
↓
Payment
↓
Database
↓
Reports / Decision Support

Benefits

- Centralized data
- Faster booking
- Better vehicle tracking
- Reduced manual work
- Improved data management
- Better decision support

Module 1: User Management

Functions

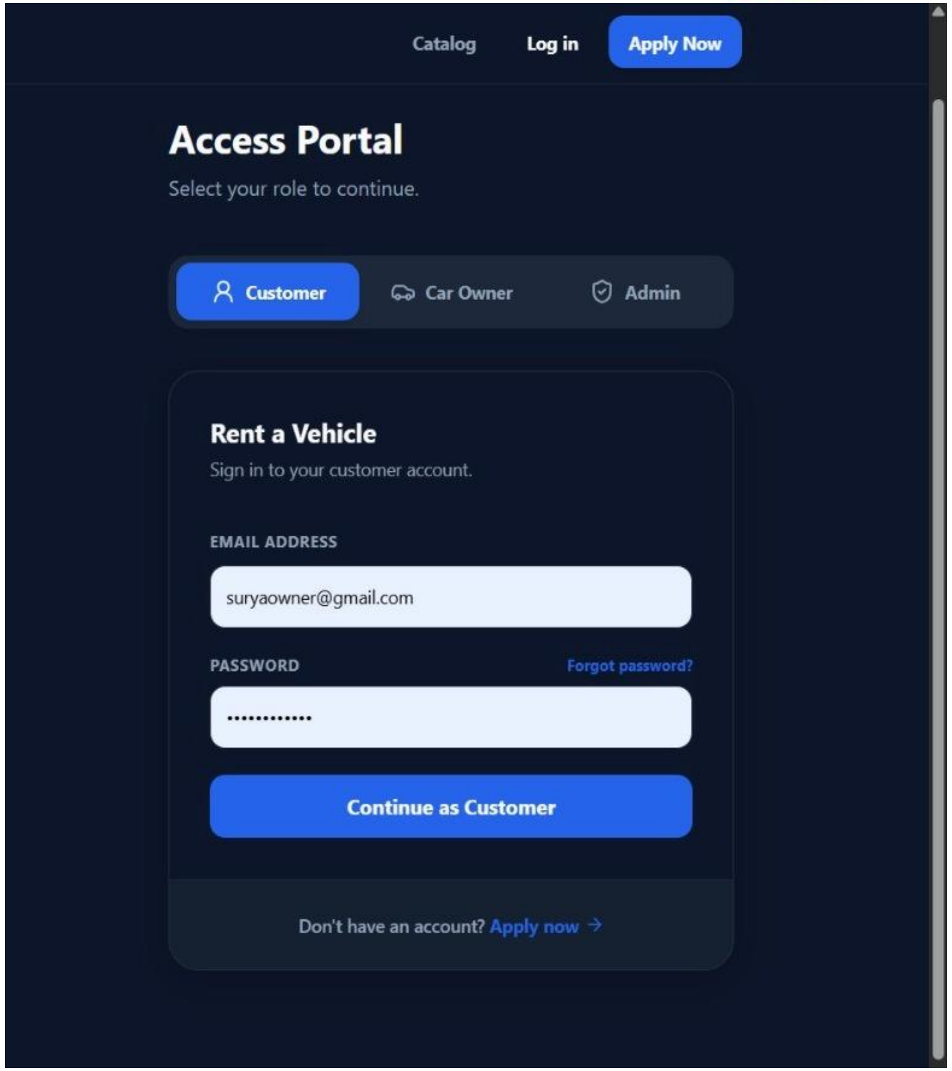
- User registration
- User login
- User authentication
- User profile management
- User information storage

Parameters

- User ID
- User Name
- Contact Details
- Login Credentials

Outcome

Secure and organized management of users.



Module 2: Vehicle Management

Functions

- Add vehicle details
- Update vehicle details
- Delete vehicle details
- Search vehicle details
- Check vehicle availability

Parameters

- Vehicle ID
- Vehicle Type
- Rental Price
- Availability

Example

Vehicle ID: V101
Vehicle Type: Sedan
Rental Price: ₹2000/day
Availability: Available

Outcome

Efficient management and tracking of rental vehicles.

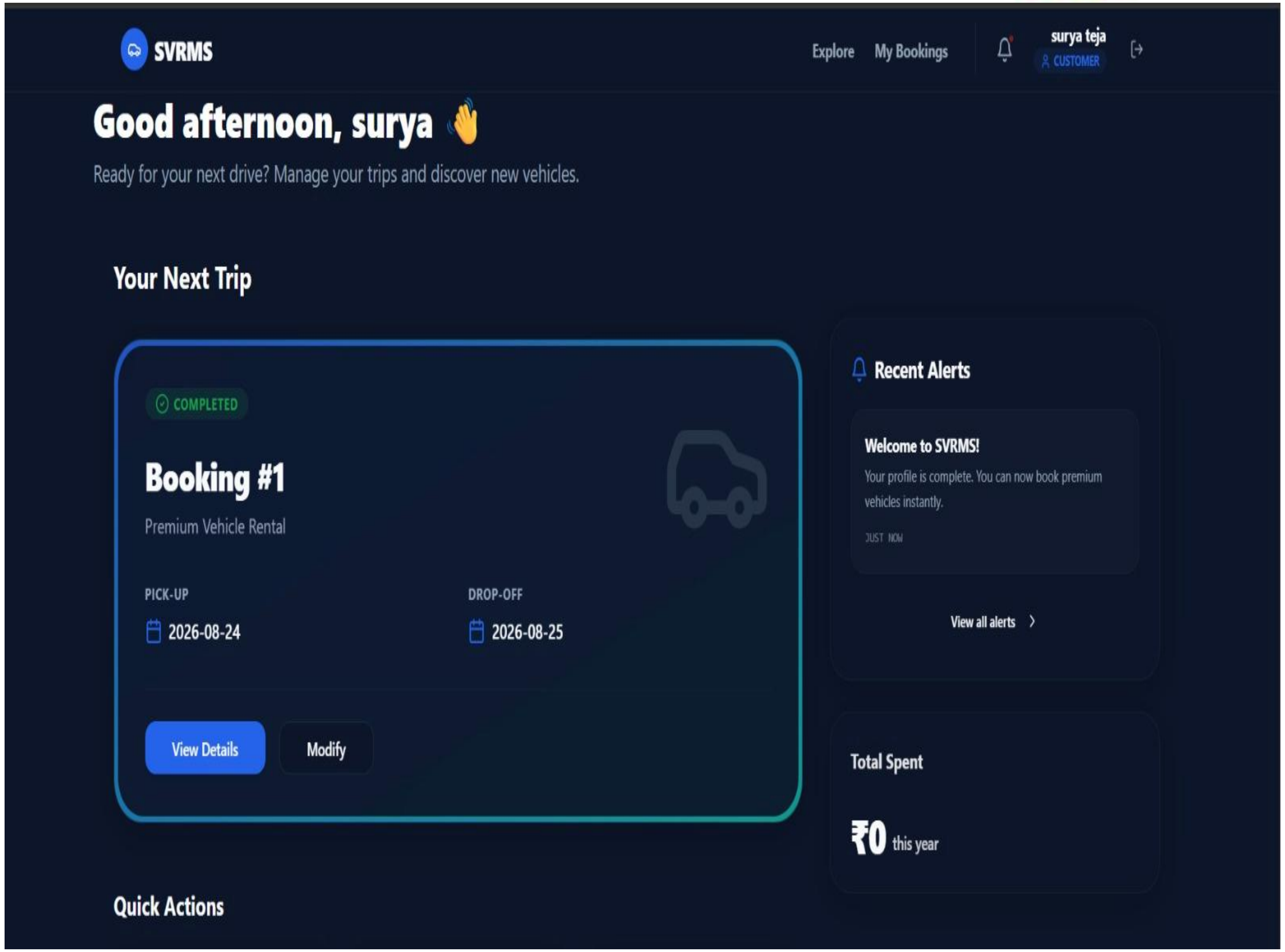
Module 3: Vehicle Booking

Functions

- Search available vehicles.
- Select a vehicle.
- Enter booking details.
- Confirm booking.
- Store booking information.
- Update vehicle availability.

Parameters

- User ID
- Vehicle ID
- Vehicle Type
- Booking Date
- Rental Price
- Availability



Module 4: Payment Management

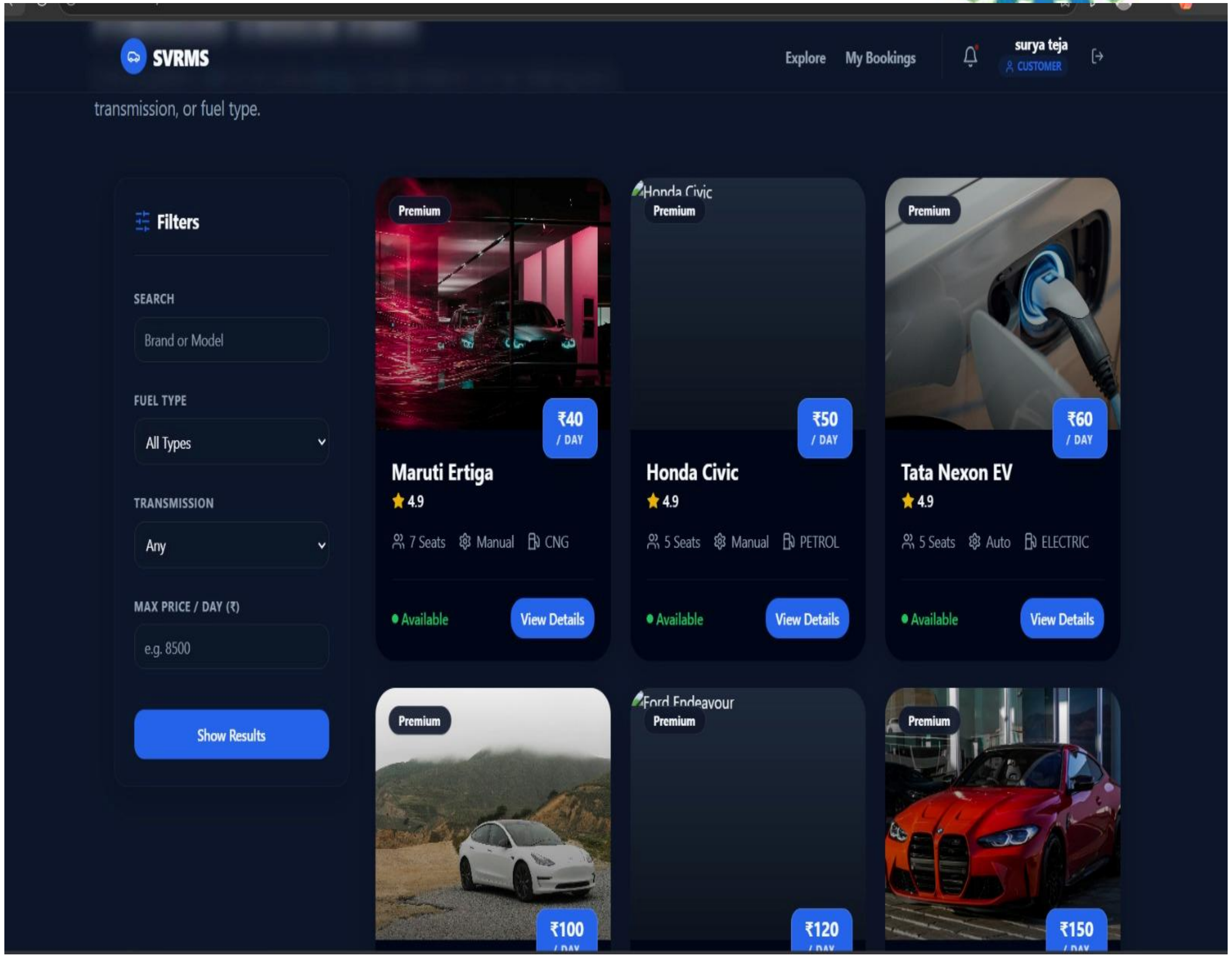
Engineer to Excel

Functions

- Calculate total rental amount.
- Store payment details.
- Validate payment.
- Update payment status.
- Maintain payment records.

Parameters

- Booking ID
- User ID
- Rental Amount
- Payment Amount
- Payment Status



CRUD Operations

The system performs CRUD operations.

C — Create

Add new users, vehicles, bookings, or payments.

R — Read

View stored information.

U — Update

Update vehicle availability, user details, booking details, etc.

D — Delete

Technologies Used

Programming Language

Java

Programming Concepts

- Java OOP
- Classes and Objects
- Encapsulation
- Methods
- Exception Handling

Database Connectivity

JDBC — Java Database Connectivity
Used to connect the Java application with the database.

Database

Relational Database
Used to store:

- Users
- Vehicles
- Bookings
- Payments

Database Operations

- SQL
- CRUD Operations
- Database Management

Advantages and Limitations

Advantages

- Easy vehicle search
- Quick booking
- Availability information
- Organized payment process

Limitations

- Requires a computer system and technical knowledge to operate.
- System failures may temporarily affect rental operations.
- Data security risks may arise if proper protection measures are not implemented.

Future Scope

The system can be enhanced with:

- Mobile application.
- GPS-based vehicle tracking.
- Online payment gateway.
- AI-based rental demand prediction.
- Dynamic rental pricing.
- Customer recommendation system.
- Predictive vehicle maintenance.
- Advanced analytics dashboard.

Conclusion

- The proposed system provides a Java-based solution for smart vehicle rental management.
- It integrates user management, vehicle management, booking, and payment management.
- JDBC enables communication between the Java application and database.
- CRUD operations provide efficient data management.
- Real-time rental information can support better operational decisions.
- The system can be further enhanced using AI, cloud technologies, GPS, and advanced analytics.

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THANK YOU